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# What is a Biotechnology Company?

# Biotechnology Companies

- A **biotechnology company** is a company whose products or services primarily use biotechnology methods for their production, design or delivery. The United Nations Convetnion on Biological Diversitty defines 'biotechnology' as: "Any technological application that uses biological systems, living organisms, or derivatives thereof, to make or modify products or processes for specific use." In other words, biotechology can be defined as the mere application of technical advances in life science to develop commercial products

- Biotechnology companies can range in size from the largest in the world such as Novo Nordisk (market cap approx $76.9B, largest independent biotech) and Genetech (purchased by Roche in 2009) with thousands of employees working on multiple product to start up operations with less than ten employees focused on one specific goal.

# Key Definitions

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- Preclinical testing - tests conducted by the company via laboratory and animal studies to show the biological activity of the compound against the targeted disease and compound is evaluated for safety. These tests take approximately 3.5 years.
- Investigational New drug Application (IND) – After completing the preclinical testing, the company files an NDA with the Food & Drug Administration (FDA) to begin to test the drug in humans. The IND becomes effective if the FDA does not reject it within 30 days. The IND shows results of previous experiments, how, where and by whom the new studies will be conducted; how the drug is thought to work in the body; any toxic effects found in animal studies; and how the compound is manufactured. In addition, the IND must be reviewed by the Institutional Review Board where the studies will be conducted. Progress reports on the clinical trials must by submitted at a minimum annually to the FDA.

# Key Definitions (continued)

- Clinical Trials - are sets of tests in medical research and drug development that generate safety and efficacy data (or more specifically, information about adverse drug reactions and adverse effects of other treatments) for health interventions (e.g., drugs, diagnostics, devices, therapy protocols). These trials are usually broken into 3 phases (on occasion a 4th Phase is also used).
- Phase I Clinical Trials –are designed to allow scientists and medical doctors to understand what effects an investigational compound has in human subjects. The goal is to study what happens to the investigational compound in the body from the time it is swallowed or injected until it is excreted, when it is excreted and how the human body reacts to the new compound from a safety and tolerability point of view. Study participants are monitored for the occurrence and severity of any side effects that they may experience.

# Key Definitions (continued)

- Phase II Clinical Trials - are designed to evaluate the safety and efficacy of an investigational compound in patients with a specific disease or condition. Phase II trials are typically conducted in a group of patients who are usually selected based upon being at the same stage of a disease. The patients are given various doses of the compound and closely monitored to compare the effects and to determine the safest dosing regimen. In many instances, multiple Phase II studies are conducted to test the compound in a variety of patient populations or indications.

- Phase III Clinical Trials - are designed to confirm the safety and efficacy of an investigational compound, and the dosage regimen chosen, in large numbers of patients with a specific disease or condition. These studies, as in the earlier phases, may involve one or more 'treatment arms', which allow the investigational drug to be compared to other available treatments, or to be tested for effectiveness in combination with other therapies. The safety and efficacy of the new compound is compared to that of the currently accepted standard treatment. Information obtained from Phase III studies is used to determine how the compound is best prescribed to patients in the future. The complete information available on the new compound is submitted to Health Authorities.

# Key Definitions (continued)

- Phase IV Clinical Trials - take place after the drug has been approved for marketing and are designed to provide broader experience in evaluating the safety and effectiveness of the new medicine in larger numbers of patients, subpopulations of patients, and to compare and/or combine it with other available treatments. These studies are designed to evaluate the long-term effects of the drug. Under these circumstances, less common adverse events may be detected.
- New Drug Application (NDA) – Following the completion of the clinical trails, the company analyzes all of the data and files this application with the FDA if the data successfully demonstrates safety and effectiveness. The NDA must contain all of the scientific information that the company has gathered. By law the FDA can take up to 6months to review the NDA.

# Key Definitions (continued)

- Fast Track Review - a process designed to facilitate the development, and expedite the review of drugs by the FDA to treat serious diseases and fill an unmet medical need. The purpose is to get important new drugs to the patient earlier. Fast Track addresses a broad range of serious diseases. If a drug receives Fast Track status, the drug becomes eligible for some or all of the following benefits : more frequent correspondence with the FDA regarding the collection of data and design of clinical trails and eligibility for Rolling Review (allows the company to submit completed sections of the NDA as they occur instead of waiting for all sections to be completed).
- Generic Drug - a drug product that is comparable to brand/reference listed drug product in dosage form, strength, route of administration, quality and performance characteristics, and intended use. It has also been defined as a term referring to any drug marketed under its chemical name without advertising.

# Key Definitions (continued)

- Abbreviated New Drug Application (ANDA) - contains data which when submitted to FDA's Center for Drug Evaluation and Research, Office of Generic Drugs, provides for the review and ultimate approval of a generic drug product. Once approved, an applicant may manufacture and market the generic drug product to provide a safe, effective, low cost alternative to the American public.
- Orphan Drugs - are drugs designed to treat people with rare diseases and infections (occurring in less than 200,000 individuals). Once the drugs are marketed to the public, orphan drug makers might not benefit from huge demand, but governments will usually subsidize many of the costs of producing these drugs.

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# Underwriting Points

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- What is the company focused on?
  - Is the company entering into a mature market with low margins?
  - Is the company working on a high profile drug (ie. cancer research, AIDS , heart disease)? High profile = high exposure
  - Will the company take the product from pre-clinical trails all the way through production or partner with a larger firm or possibly sell the technology?
- What does the company's pipeline look like?
  - How many projects are they working on?
  - What phase of clinical trials are they in?
  - Do they have any products on the market?
  - Do the products have off-label applications?

# Underwriting Points (continued)

- What is the company's financial condition?
  - Liquidity is paramount (especially for small companies)
  - Current Ratio and Working Capital are very important
  - How much cash do they have on hand and how is it held?
  - Debt is usually low to non-existant
  - Compare cash burn to cash balance
  - Focus on the company's R&D spending
  - What sources of cash can the company draw from to fund operations (LOC's, equity investment, partnerships, milestone payments)
- What is the structure of the company's Board of Directors?
  - Members usually have a scientific background
  - Previous public company experience is favorable
  - Previous experience with the FDA is a large plus

# Underwriting Points (continued)

- Review the company's stock chart
  - What moves the company's stock (both price and volume)?
    - Drug pipeline developments
    - Patent expirations
    - Competitors actions
    - Regulatory environment changes
    - Outside investment
- How is the company's technology valued?
  - A rough idea can be gained by Market Cap – Cash + Investments
  - What is the value of the addressable market of the technology

# Other Considerations

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- Due to the volatile nature of this industry limits management is key
- Pricing on these accounts varies due to several factors such as clinical stage and financial viability
- Biotech companies are not immune to accounting issues:
  - accounting for R&D costs should be watched (some companies have been caught capitalizing R&D costs onto their balance sheet which would artificially inflate earnings
  - accounting for stock based compensation
  - how are licensing agreements recognized
  - revenue recognition regarding wholesaler agreements
- Due to the varying sizes of company's in this industry M&A activity does pose a risk for underwriting, especially in the primary layer
- Product liability issues

# Other Considerations (continued)

- Insurance company and Medicare/Medicaid reimbursement issues should be carefully reviewed
- Patents expirations and generic competition
- Sudden and unexpected increases in raw materials prices may not be transferrable to the customers

# Matrixx Decision

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- Background :
  - In re Carter-Wallace Inc, Securities Litigation (2nd Cir 1988) – bright line rule under which the failure to disclose reports of adverse drug effects will not be deemed material unless the reports are statistically significant
  - April 2004 shareholders brought a securities suite against Matrixx claiming the company had failed to disclose a possible link between their product Zicam (otc cold remedy) and a condition called anosmia
  - The lower court relyed on Carter-Wallace and dismissed the claim sighting the lack of statistical significance; also the court ruled that the plaintiffs had failed to prove a strong inference of scienter
  - On appeal the 9th Circuit reversed the decision; the court rejected the statistical significance standard ; also the court stated that scienter can be raised as the company's actions represented an "extreme departure from the standards of ordinary care"
- The Supreme Court Ruling
  - March 22, 2011 the Supreme Court unanimously affirms the 9th Circuits decision
  - The court stated that medical professionals and the FDA can and do rely on evidence that may not be statistically significant to asses the risk and take regulatory action
  - Regarding the scienter issue : the court again affirmed the 9th Circuit, stating that the company omitted the information not because of statistical significance but because of the effect the information would have on the market

# Matrixx Decision (continued)

- Conclusion
  - The Supreme Court's ruling will make it more difficult to obtain a dismissal of a securities claim based on the decisions of the company as to what adverse effects of a product should be disclosed

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